A Primer for Students

Metrics, field equations, and the first exact solutions

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Role of AI.

This primer is based on an earlier draft of mine. Claude (Anthropic) edited, corrected, and condensed it, and checked its formulas with SymPy (Python).

1. What this primer is for

The notebooks in this repository solve Einstein's field equations step by step, and the Introduction and Summary explains what the solutions mean. Both assume that the reader already knows what a metric is and what the field equations say. This primer supplies that background for a student who knows calculus and a little linear algebra but has not yet studied general relativity. It ends with two worked examples, Schwarzschild–de Sitter and the expanding universe, which use the same method as all the notebooks.

2. Distance and the metric

In the xy-plane, the distance d between two points follows from the Pythagorean theorem, $d^2 = \Delta x^2 + \Delta y^2$, and in three dimensions $d^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$. The rule that turns coordinate differences into a distance is called the metric.

In special relativity, time joins the three space coordinates, and the corresponding quantity is the spacetime interval

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

The time term has the opposite sign from the space terms. That sign difference is what makes spacetime different from four-dimensional space: for a light ray $\Delta s^2 = 0$, for a clock moving slower than light $\Delta s^2 > 0$, and $\Delta s/c$ is the time the clock itself measures. Writing the signs as (+, −, −, −), as here and in the notebooks, is one of two common conventions; many books use (−, +, +, +) instead, which flips every sign but changes no physics.

Indices.

It is customary to number the coordinates $x^0 = t$, x^1, x^2, x^3, with Greek indices μ, ν running from 0 to 3. For two neighboring points the interval becomes the line element

$$ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu,$$

where, by Einstein's summation convention, a repeated index is summed over all its values. The coefficients $g_{\mu\nu}$ are the components of the metric tensor. Because $g_{\mu\nu} = g_{\nu\mu}$, a 4-dimensional metric has 10 independent components (in N dimensions, $N(N + 1)/2$).

Even flat spacetime can have non-constant components. In spherical coordinates (t, r, θ, φ),

$$ds^2 = c^2\,dt^2 - dr^2 - r^2\,d\theta^2 - r^2\sin^2\theta\,d\varphi^2,$$

so $g_{00} = c^2$, $g_{11} = -1$, $g_{22} = -r^2$, $g_{33} = -r^2\sin^2\theta$, and all others are zero. Gravity shows up when no change of coordinates can make the components constant: then spacetime is curved.

A note on units: the notebooks use t as the time coordinate, so c^2 appears in g_{00}. Many books use $x^0 = ct$ instead, or units with $c = 1$. These are bookkeeping choices only.

3. The field equations

Curvature is measured by the Ricci tensor $R_{\mu\nu}$, which is built from first and second derivatives of the metric. The notebooks compute it in three stages: the connection coefficients of the first kind, $\Gamma_{\delta\nu\gamma} = \tfrac{1}{2}(\partial_\nu g_{\delta\gamma} + \partial_\gamma g_{\nu\delta} - \partial_\delta g_{\nu\gamma})$; those of the second kind, $\Gamma^\mu{}_{\nu\gamma} = g^{\mu\delta}\Gamma_{\delta\nu\gamma}$; and then $R_{\mu\nu}$ from sums of their derivatives and products.

Einstein's equations relate curvature to energy and momentum. They are usually written

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} + \Lambda\,g_{\mu\nu} = \kappa\,T_{\mu\nu} \quad \text{(textbook form)},$$

where $R = g^{\mu\nu}R_{\mu\nu}$ is the curvature scalar, $T_{\mu\nu}$ is the stress-energy tensor of matter and radiation, $\kappa = 8\pi G/c^4$, and Λ is the cosmological constant (dark energy). The overall signs depend on the conventions a book uses.

Taking the trace (multiplying by $g^{\mu\nu}$ and summing, with $g^{\mu\nu}g_{\mu\nu} = D$ in D dimensions) eliminates R, and the equations can be rewritten with $R_{\mu\nu}$ alone on the left. In the notebooks' conventions the result is

$$R_{\mu\nu} = \Lambda\,g_{\mu\nu} - \kappa\,[\,T_{\mu\nu} - T\,g_{\mu\nu}/(D-2)\,], \quad T = g^{\mu\nu}T_{\mu\nu}.$$

In 4D the factor $1/(D-2)$ is the familiar ½. A point worth noticing: Λ moves from the left side to the right side with a plus sign, because the trace contributes $2\Lambda g_{\mu\nu}$ in 4D, outweighing the $-\Lambda g_{\mu\nu}$ one might expect. (In D dimensions, the Λ in this form equals $2\Lambda_{\text{std}}/(D-2)$, where Λ_{std} is the constant in the textbook form; the two agree in 4D.)

In empty space $T_{\mu\nu} = 0$, which gives the Λ-vacuum equations solved in all the notebooks:

$$R_{\mu\nu} = \Lambda\,g_{\mu\nu}.$$

These are $D(D+1)/2$ coupled, second-order partial differential equations for the metric components. Assumptions such as spherical symmetry reduce them to ordinary differential equations in r or t, which is what makes exact solution possible.

Why Λ must be constant.

One might ask whether Λ could vary in time or space. It cannot. The Bianchi identity, a property of every metric, says that the combination $R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu}$ has zero divergence. In vacuum, $R_{\mu\nu} = \Lambda g_{\mu\nu}$ gives $R = D\Lambda$, so $R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = (1 - D/2)\Lambda\,g_{\mu\nu}$. Since the metric itself has zero covariant derivative, the divergence is $(1 - D/2)\partial_\nu\Lambda$, and setting it to zero forces $\partial_\nu\Lambda = 0$ for $D > 2$. Every attempt in the notebooks to let Λ depend on t failed for exactly this reason.

4. How the notebooks solve the equations

Every case follows the same recipe:

- Choose a line element with unknown functions, for example A(t, r), B(r), C(r), guided by the symmetry of the situation.
- Compute the connection coefficients and $R_{\mu\nu}$, and write out each field equation.
- Look for an equation, or a combination of equations, that can be solved for one unknown.
- Substitute the result back into the remaining equations and repeat until every equation becomes an identity, $\Lambda = \Lambda$.

The two examples below show this recipe on the simplest cases.

5. Worked example: Schwarzschild–de Sitter

(Notebook 4D_b.) Schwarzschild's 1916 solution, the first exact solution of Einstein's equations, describes the spacetime outside a spherical mass. With Λ included, start from

$$ds^2 = A(t, r)\, dt^2 - B(r)\, dr^2 - r^2\, d\theta^2 - r^2 \sin^2\theta\, d\varphi^2.$$

The notebook finds four field equations, of which R_{33} repeats R_{22}. The combination $R_{00} - R_{11} - 2R_{22}$ contains only B:

$$1/r^2 - 1/(r^2 B) + B'/(rB^2) = \Lambda.$$

Writing $u = r/B$, this becomes $1 - u' = \Lambda r^2$, so $u = r - \alpha - \Lambda r^3/3$ with a constant of integration α. Hence

$$B(r) = 1 \,/\, (1 - \alpha/r - \Lambda r^2/3).$$

With this B, the R_{22} equation reduces to $(AB)' = 0$, so $A(t, r) = \mathcal{A}(t)/B(r)$ for an arbitrary function $\mathcal{A}(t)$. The free $\mathcal{A}(t)$ is not new physics: replacing t by a new time coordinate τ with $c\, d\tau = \sqrt{\mathcal{A}}\, dt$ removes it. (That every spherically symmetric vacuum solution is static in this sense is Birkhoff's theorem.) Choosing $\mathcal{A} = c^2$ and $\alpha = 2GM/c^2$ gives the Schwarzschild–de Sitter solution:

$$ds^2 = c^2 f\, dt^2 - dr^2/f - r^2\, d\theta^2 - r^2 \sin^2\theta\, d\varphi^2, \quad f = 1 - 2GM/(c^2 r) - \Lambda r^2/3.$$

- **Schwarzschild ($\Lambda = 0$):** $f = 1 - 2GM/(c^2 r)$. The factor f in g_{00} makes clocks near the mass run slow; $f = 0$ at the Schwarzschild radius $r = 2GM/c^2$, the event horizon of a black hole.
- **de Sitter ($M = 0, \Lambda > 0$):** $f = 1 - \Lambda r^2/3$, empty space with dark energy; $f = 0$ at the cosmological horizon $r = \sqrt{(3/\Lambda)}$.
- **Anti-de Sitter ($M = 0, \Lambda < 0$):** $f = 1 + |\Lambda| r^2/3$.

6. Worked example: the expanding universe

(Notebook 4D.) For a universe that is the same everywhere, start from

$$ds^2 = A(t)\, dt^2 - a(t)^2\, [\, B(r)\, dr^2 + C(r)\, r^2\, (d\theta^2 + \sin^2\theta\, d\varphi^2)\,].$$

The scale factor a(t) multiplies every spatial distance: a growing a(t) means an expanding universe, $\dot{a} > 0$ means expansion, and $\ddot{a} > 0$ means accelerating expansion. Only a^2 appears in the metric, so a is taken positive.

The R_{00} equation, $3\ddot{a}/(aA) - 3\dot{a}\dot{A}/(2aA^2) = \Lambda$, integrates to

$$A(t) = 3\dot{a}^2 / (\Lambda a^2 - 3k),$$

where k is a constant of integration, the curvature constant. The spatial equations then give $B = 1/(1 - kr^2)$ when $C = 1$, the familiar FLRW form, valid for $kr^2 < 1$. Requiring $A(t) = c^2$ (so that t is the time measured by observers moving with the expansion) gives

$$\dot{a}^2 = \Lambda c^2 a^2/3 - kc^2,$$

which is the Friedmann equation for empty space with dark energy. It can be solved by hand by separating variables:

Case	Scale factor a(t)	Behavior
$k > 0, \Lambda > 0$	$\sqrt{(3k/\Lambda)}\cosh(\sqrt{(\Lambda/3)}\,ct)$	contracts, bounces, re-expands
$k < 0, \Lambda > 0$	$\sqrt{(3\lvert k\rvert/\Lambda)}\sinh(\sqrt{(\Lambda/3)}\,ct)$	expands from a = 0, then accelerates
$k < 0, \Lambda = 0$	$\sqrt{\lvert k\rvert}\,ct$	linear growth (the Milne universe)
$k = 0, \Lambda \geq 0$	$\alpha \exp(\pm\sqrt{(\Lambda/3)}\,ct)$	exponential (de Sitter); constant if $\Lambda = 0$
$k < 0, \Lambda < 0$	$\sqrt{(3\lvert k\rvert/\lvert\Lambda\rvert)}\sin(\sqrt{(\lvert\Lambda\rvert/3)}\,ct)$	expands, then recollapses

The $k = 0, \Lambda > 0$ row (exponential growth, the de Sitter universe) is the same spacetime as the de Sitter case of Section 5, described in different coordinates: one in which space expands, and one in which it is static. Learning to recognize one spacetime in several coordinate systems is a recurring theme in the notebooks.

Adding matter and radiation.

With a perfect fluid of density ρ and pressure p on the right-hand side of the field equations, the same calculation gives the full Friedmann equations

$$(\dot{a}/a)^2 = (8\pi G/3)\rho + \Lambda c^2/3 - kc^2/a^2, \qquad \ddot{a}/a = -(4\pi G/3)(\rho + 3p/c^2) + \Lambda c^2/3.$$

Matter and radiation slow the expansion, and Λ speeds it up. For a flat universe ($k = 0$):

Dominant content	Scale factor
Radiation	$a \propto t^{1/2}$
Matter	$a \propto t^{2/3}$
Dark energy (Λ)	$a \propto \exp(Ht)$, $H = c\sqrt{(\Lambda/3)}$
Matter + Λ	$a \propto \sinh^{2/3}(3Ht/2)$

The vacuum solutions in the notebooks describe the last stage, when dark energy dominates. Our universe is already about 70% dark energy, so this is its far future.

7. Where to go next

- Read the Introduction and Summary, which assumes the background of this primer, and its Glossary.
- Work through the notebooks in order: 4D, 5D, 6D_1, 6D_2, 7D_1, 7D_2, 7D_3. Each adds one new idea, such as curled dimensions, the Kaluza–Klein bubble, or the curvature test.
- For a textbook treatment, J. Foster and J. D. Nightingale, A Short Course in General Relativity, 3rd ed., Springer (2006), is the book I started from.